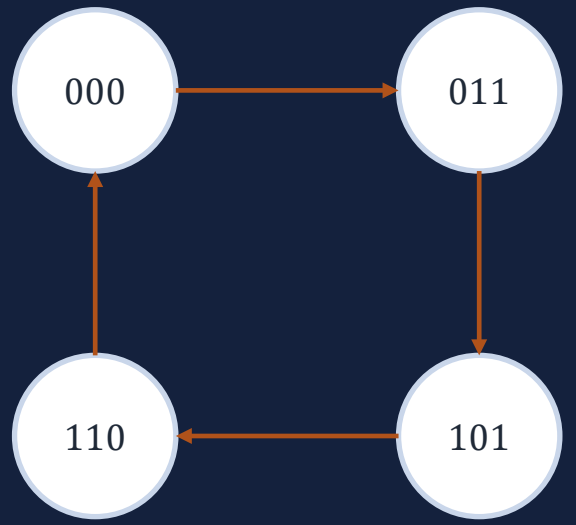
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 29

Comprehensive Final Review



One worked problem per unit · numbers to memory · sequential design and unused states · mixed practice set · exam strategy

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapters 1–7 (comprehensive review)

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Today’s roadmap

1 §3.3

Foundations

Numbers, algebra, K-maps

2 §4.11

Combinational

Blocks and design

3 §5.8

Sequential

Design, counters

4 §7.3

Memory

Expansion, Hamming

By the end of class you can…

- Do signed arithmetic and base conversion quickly
- Minimize with don’t-cares and implement on a MUX
- Design a counter and fix an unused-state lockout
- Trace shift-register counters
- Expand memories and correct single-bit errors

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Agenda for today’s session



Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; how the final is organised	—
0:05 – 0:25	Numbers, K-maps	Reviews 1, 2
0:25 – 0:40	Multiplexer implementation	Review 3
0:40 – 0:50	Break	
0:50 – 1:20	Sequential design; counters	Reviews 4, 5
1:20 – 1:30	Expansion and Hamming	Reviews 6, 7
1:30 – 1:50	Mixed practice set; strategy	Practice set

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Review 1 — Signed arithmetic

§1.6

REVIEW Compute $(-37) + (-58)$ in 8-bit 2's complement. Overflow? Then convert $(3B)_{16}$ to binary and decimal.

−37	1	1	0	1	1	0	1	1
−58	1	1	0	0	0	1	1	0
sum	1	1	0	1	0	0	0	0	1

The end carry (highlighted) is dropped.

Carry into and out of the sign bit are both 1: $V = 0$.

1. $37 = 00100101$; invert 11011010, add 1: $-37 =$ **11011011**
2. $58 = 00111010$; invert 11000101, add 1: $-58 =$ **11000110**
3. Add: $1\ 10100001$; drop the end carry
4. Sign 1: negative. Invert 01011110, add 1: $01011111 = 95$
5. So the sum is **−95** ✓
6. Carries into and out of the sign are both 1: $V = 0$
7. $(3B)_{16}$: $3 = 0011$, $B = 1011$ → **0011 1011**
8. $3 \cdot 16 + 11 =$ **59**

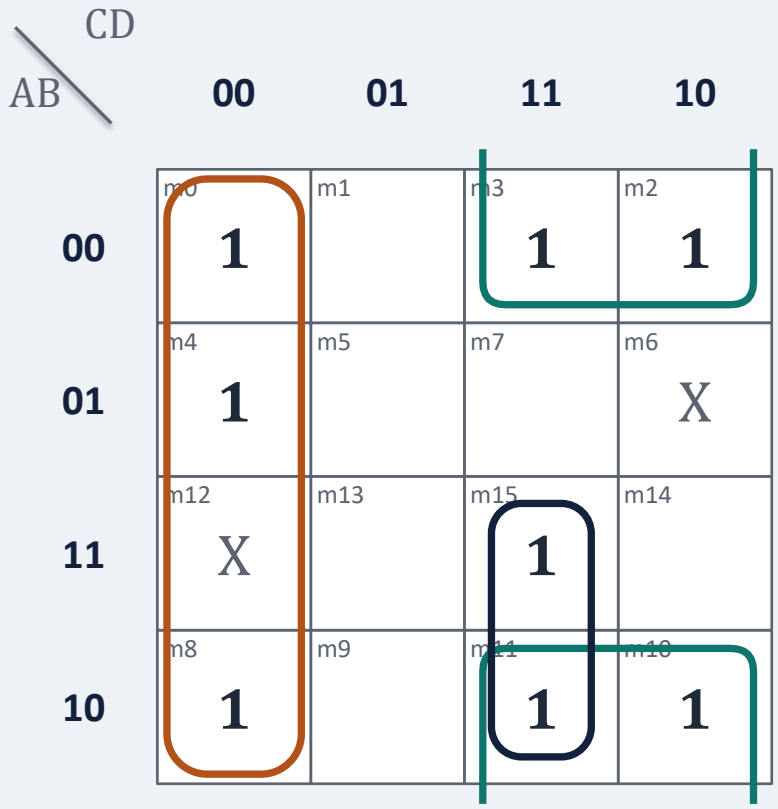
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Review 2 — K-map with don’t-cares

§3.5

REVIEW

Minimize $F = \Sigma m(0,2,3,4,8,10,11,15) + d(6,12)$.



1. Place 1s at 0, 2, 3, 4, 8, 10, 11, 15; X at 6, 12
2. Column $CD = 00$: m0, m4, m8 and X at m12 → $C'D'$
3. Quad m2, m3, m10, m11: $B = 0$, $C = 1$ → $B'C$
4. m15 is left: pair with m11 (1011, 1111) → ACD
5. X at m6 would not enlarge a needed group: leave it 0
6. Every 1 is covered: $F = ACD + C'D' + B'C$

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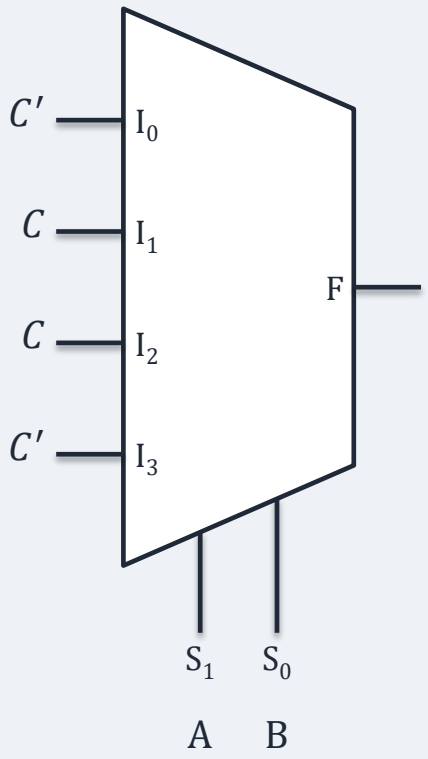
Review 3 — Function on a MUX

§4.11

REVIEW

Implement $F(A,B,C) = \Sigma m(0,3,5,6)$ on a 4-to-1 MUX with A, B on the selects. What function is it?

A B	C=0	C=1	Input
00	1	0	I0 = C'
01	0	1	I1 = C
10	0	1	I2 = C
11	1	0	I3 = C'



1. For each AB: F at $C = 0$, then $C = 1$
2. $AB = 00$: m0 = 1, m1 = 0 → C'
3. $AB = 01$: m2 = 0, m3 = 1 → C
4. $AB = 10$: m4 = 0, m5 = 1 → C
5. $AB = 11$: m6 = 1, m7 = 0 → C'
6. 0, 3, 5, 6 each have an even number of 1s
7. $F = (A \oplus B \oplus C)'$: even parity

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Break

10 minutes — we resume at the 0:50 mark

Next: sequential design, counters and memory

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Review 4 — Design a counter

§5.8

REVIEW

Design a counter that repeats 000 → 011 → 101 → 110 with D flip-flops. Use the unused states as don't-cares.

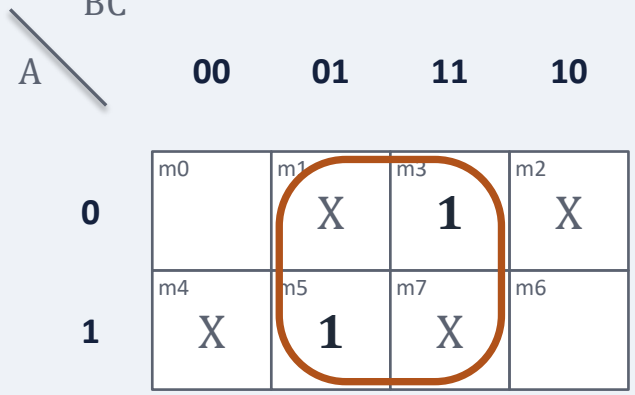
ABC	next	DA DB DC
000	011	0 1 1
011	101	1 0 1
101	110	1 1 0
110	000	0 0 0
001	X	X X X
010	X	X X X
100	X	X X X
111	X	X X X

1. Four states, three flip-flops; unused 001, 010, 100, 111 → X
2. D flip-flops: D = next-state bit
3. Next A = 0, 1, 1, 0 for 000, 011, 101, 110: equals C → $D_A = C$
4. Next B = 1, 0, 1, 0: the opposite of B → $D_B = B'$
5. Next C = 1, 1, 0, 0: the opposite of A → $D_C = A'$
6. No gates at all besides one inverter per bit
7. Next: check where the unused states go

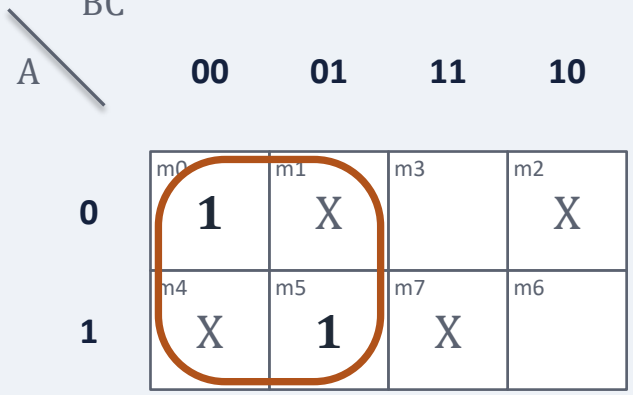
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Review 4 — The three K-maps

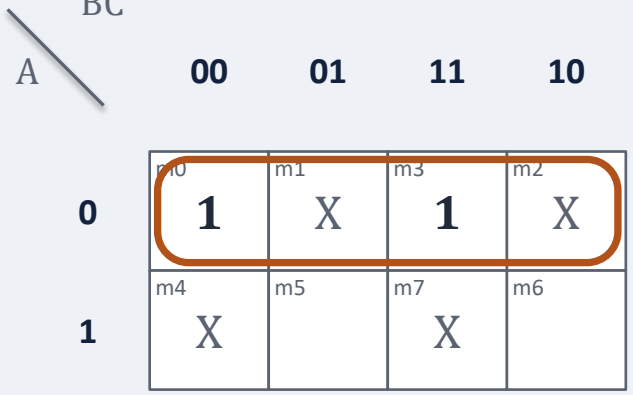
§5.8



$D_A = C$



$D_B = B'$



$D_C = A'$

1. Next $A = 1$: m3 (→101), m5 (→110)
2. With X at m1, m7: quad $C = 1$
3. Next $B = 1$: m0 (→011), m5 (→110)
4. With X at m1, m4: quad $B = 0$
5. Next $C = 1$: m0 (→011), m3 (→101)
6. With X at m1, m2: quad $A = 0$

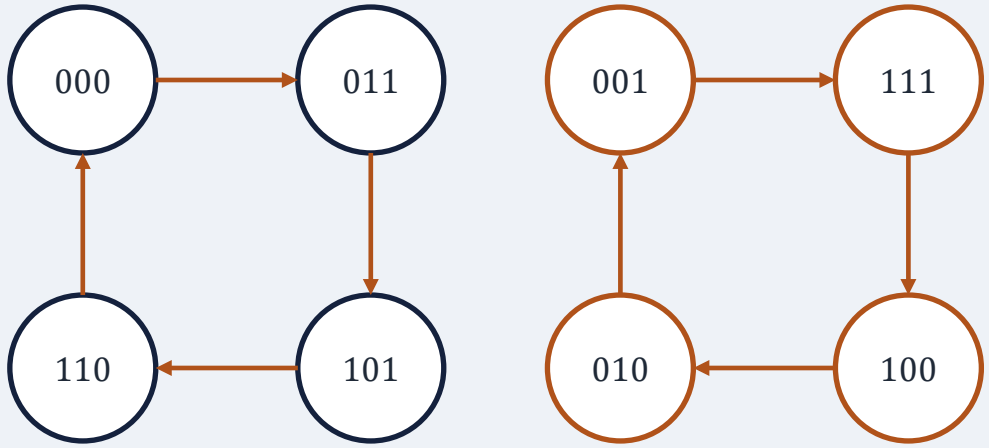
Each X was set to make the biggest group. Those choices decide where 1, 2, 4 and 7 go: next slide.

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Review 4 — Unused states lock out

§5.8

With $D_C = A'$: a second cycle



Left: the count. Right: 001 → 111 → 100 → 010 → 001, which never returns.

e.g. 001: $D_A = C = 1, D_B = B' = 1, D_C = A' = 1 \rightarrow 111$

Power up in 001 and the counter is stuck forever.

The fix

Choose the don't-cares so every unused state leads into the count.

$D_C = A' + BC$ (one more AND gate) sends

001 → 111 → 101, 010 → 001, 100 → 010

e.g. 111: $D_A = 1, D_B = 0, D_C = 0 + 1 = 1 \rightarrow 101$

The count is unchanged: $BC = 0$ on 101 and 110, $A' = 1$ on 000 and 011

Every unused state now reaches the count within 4 clocks: self-correcting.

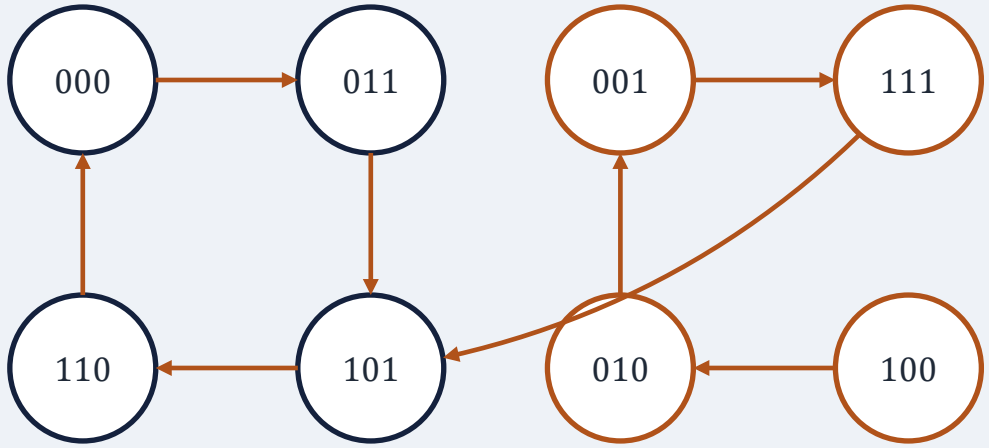
Found by trying every assignment of the unused next states and keeping the cheapest that recovers.

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Review 4 — The corrected counter

§5.8

With $D_C = A' + BC$: every state recovers



Unused	next	path to the count
100	010	100 → 010 → 001 → 111 → 101
010	001	010 → 001 → 111 → 101
001	111	001 → 111 → 101
111	101	111 → 101

1. Follow each unused state with the new equations
2. 100 → 010 → 001 → 111 → 101: longest, **4 clocks**
3. Cost: one AND gate and one OR gate

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Review 5 — A Johnson counter

§6.5

REVIEW

A 3-bit Johnson counter starts at 000. List its states and decode state 110.

Clock	$Q_2Q_1Q_0$
0	000
1	100
2	110
3	111
4	011
5	001

1. Shift right each clock; Q_0' feeds back into Q_2
2. States: 000 → 100 → 110 → 111 → 011 → 001
3. Six states ($2n$ for $n = 3$), then back to 000
4. Each state has a unique pair of adjacent bits
5. 110 is the only state with $Q_1 = 1$, $Q_0 = 0$
6. Decode 110 with $Q_1 Q_0'$ (one 2-input AND)

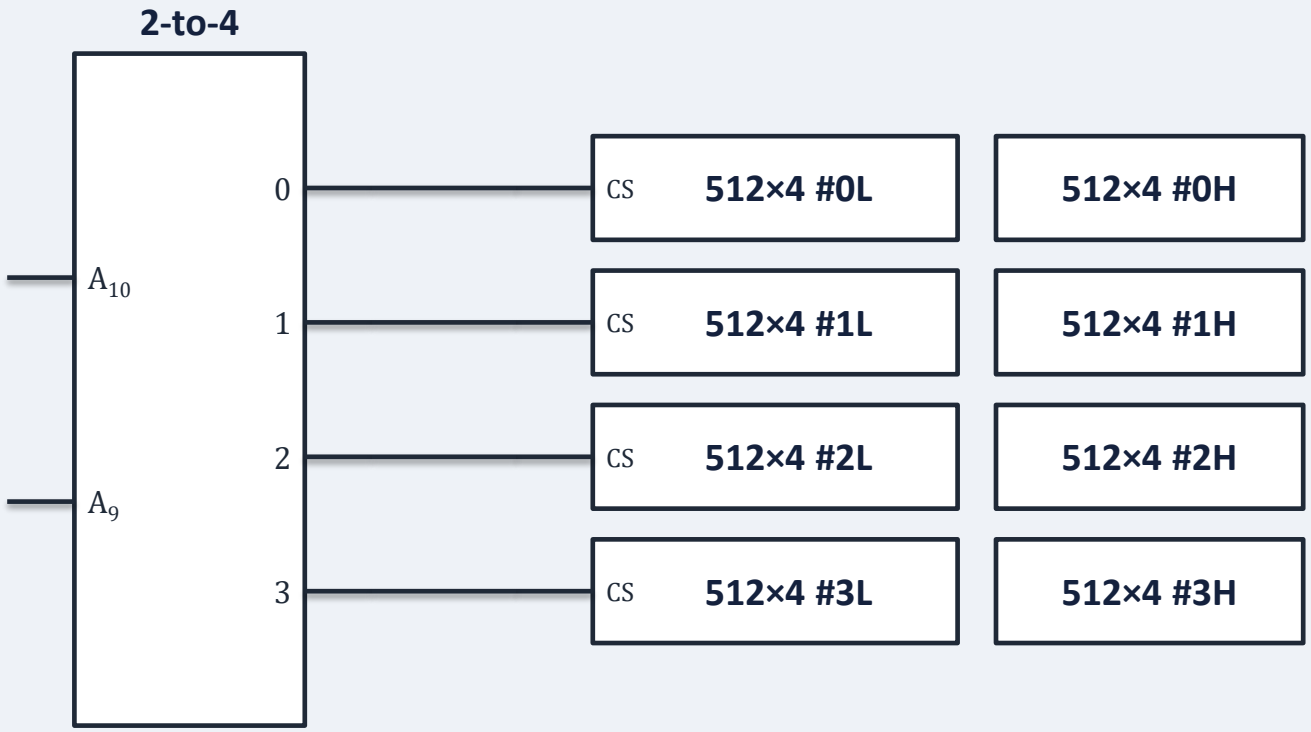
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Review 6 — Memory expansion

§7.3

REVIEW

Build 2K × 8 from 512 × 4 chips. Which bank holds address 0x4A0?



Each high chip shares its bank's chip select; $A_8 \ldots A_0$ go to every chip.

1. Width: $8/4 = 2$ chips per bank
2. Depth: $2048/512 = 4$ banks → **8 chips**
3. $2K = 2^{11}$: 11 lines; $512 = 2^9$: A_8–A_0 to every chip
4. $A_{10}A_9$ → the 2-to-4 decoder picks the bank
5. $4A0_{16} = 100\ 1010\ 0000$ (11 bits)
6. $A_{10}A_9 = 10$ → **bank 2**

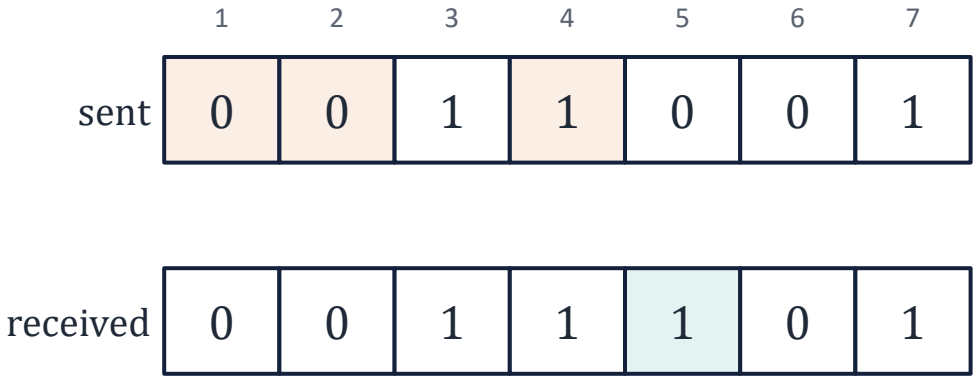
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Review 7 — Hamming (7,4)

§7.4

REVIEW

Encode 1001. The word 0011101 is received: correct it.



1. Data 1, 0, 0, 1 at positions 3, 5, 6, 7
2. $P_1 = 1 \oplus 0 \oplus 1 = 0;\ P_2 = 1 \oplus 0 \oplus 1 = 0$
3. $P_4 = 0 \oplus 0 \oplus 1 = 1 \rightarrow$ **0011001**
4. Received 0011101: 1s at 3, 4, 5, 7
5. XOR: $011 \oplus 100 \oplus 101 \oplus 111 = 101 = 5$: bit 5 is wrong
6. Flip it → **0011001** ✓

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Mixed practice set — 15 minutes

§2.4

YOUR TURN

≈ 15 min

1. Multiply 1011 × 0110 in binary. Check in decimal.
2. Minimize $F(A, B, C) = \Sigma m(1,3,5,7)$.
3. $t_{pd} = 5$, $t_{comb} = 12$, $t_{setup} = 3$ ns: f_{max}?
4. What does a JK flip-flop do with $J = K = 1$?
5. How many chips of 1K × 4 make 4K × 8?
6. Which PLD shares product terms between outputs?

Work alone.

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Mixed practice set — Solutions

§2.4

SOLUTION

1. 1. 1011×0110: multiplier bits 1 and 2 are 1

2. Partial products 10110 and 101100; add: **1000010**

3. $66 = 11 \times 6$ ✓

4. 2. 1, 3, 5, 7 are every row with $C = 1$: a quad → $F = C$

5. 3. $T \geq 5 + 12 + 3 = 20$ ns → **50 MHz**

6. 4. $J = K = 1$: it **toggles** ($Q^+ = Q'$)

7. 5. Width $8/4 = 2$, depth $4K/1K = 4$ → **8 chips**

8. 6. The **PLA**: its OR array is programmable,

9. so one product term can feed several outputs

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Final exam strategy

During the exam

Skim everything first; start with what you know.

Name the procedure and number the steps: partial credit follows structure.

State every assumption: MSB, state assignment, active levels.

Check each answer on one row or by converting back.

Bring

Pencil, eraser, calculator.

Know by heart

DeMorgan, consensus, $x + x'y = x + y$

Gray-code map labels; the full adder

$Q^+ = D, JQ' + K'Q, T \oplus Q$ and the excitation tables

$T \geq t_{pd} + t_{comb} + t_{setup}$

$2^k \times n$ memories; $2^k \geq n + k + 1$

The Key takeaways slide of every lecture is the checklist.

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Key takeaways

Numbers

2's complement, $V = C_{in} \oplus C_{out}$

Minimize

K-map, tabular, don't-cares

Combinational

decoders, MUXes, adders

Sequential

diagram → table → maps → check

Counters

check unused states

Memory

$2^k \times n$, Hamming

Every digital system is registers joined by combinational logic.

Next: the final exam — good luck

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